

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`
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- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-ae8541ac7a51732f6.jsonl`
- SHA-256: `6789bb152373af988df23882910fc15dd383a15949c95fcbc4e84f9bba29a5aa`
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```

Transcript

1. user (2026-06-12T14:41:18.760Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).
Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [medium] 'PB Vision x PodPlay template from the 2026-06-10 audit' is unresolvable from the repo

The PIVOT B row (docs/PMF_FIELD_SIGNALS.md line 70) tells the reader to follow 'the PB Vision x PodPlay template from the 2026-06-10 audit'. Neither 'PB Vision' nor 'PodPlay' appears anywhere else in the repo ? including docs/PMF_MARKET_RESEARCH.md section 4 (the competitor/pricing table) and docs/COMMERCIALIZATION.md. The reference is real (it is recorded in session memory, memory/current-state.md lines 388/394: PB Vision \$99/yr B2C + \$8/hr-of-video partner API, sold through whoever owns the facility relationship), so it is not fabricated provenance ? but a repo reader who lands on PIVOT B has no way to discover what the template is, and the doc that PMF_FIELD_SIGNALS names as 'the hypotheses under test' (PMF_MARKET_RESEARCH.md) does not contain it.

Proposed fix: Add a one-line footnote in PMF_FIELD_SIGNALS.md describing the template (PB Vision: \$99/yr consumer + \$8/hr-of-video partner/API motion through facility owners ? e.g. PodPlay), and/or add an audit-addendum row to PMF_MARKET_RESEARCH.md section 4 so the citation resolves in-repo.

2. user (2026-06-12T14:41:25.231Z)

docs/pmf-field-signals

3. user (2026-06-12T14:41:25.935Z)

1 # C13 Field Signals ? what academy visits must tell us (and when to change course)

```

2
3  **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4  (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5  visiting academies and reading signals on the ground*).
6  **Companions:** [ `FIELD_RESEARCH_ASSOCIATE_FLYER.md` ](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7  (the recruit hand-out: script + ground rules) .
8  [ `FIELD_VISIT_ANSWER_SHEET.md` ](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9  [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11  **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12  first visit** ? what those answers *mean* and what totals trigger which decision.
13  Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14  15 ambiguous conversations as whatever we hoped to hear.
15
16  ---
17
18  ## 1. The hypotheses under test (each maps to script questions)
19
20  | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21  |---|---|---|---|
22  | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23  | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24  | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25  | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26  | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27  | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BY0 multi-court
footage) | multi-court video already useful to them | Q6 |
28  | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30  Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31  hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32  roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
33  break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35  ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37  Score each hypothesis on the answer sheet **from what they said and showed, not
38  what you inferred**. When between two grades, take the lower one.
39
40  | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41  |---|---|---|---|
42  | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43  | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
44  | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
45  | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BY0 footage, price) | uses a competitor, mildly satisfied | competitor solves it,
satisfied, sticky |
46  | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their
own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free"
|
47  | **H6 multi-court** | complains about it in their own words before/without prompting |
agrees when asked | multi-court footage already works for them |
48  | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |
49
50  **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral*

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51 signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2
52 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a
53 polite conversation, not a customer.
54
55 **Disconfirming-evidence quota:** every visit sheet must record at least one
56 verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a
57 sheet has only positives, the visit gets re-done or discounted ? per the flyer's
58 ground rule 4, a reasoned "no" is worth more than five vague yeses.
59
60 ## 3. Pre-registered decision rules (n = 10?15 academies)
61
62 Counts, not percentages ? at this sample size we're reading direction, not
63 estimating proportions. Evaluate after **every 5 completed sheets** ; final call at
64 the agreed n. "GREEN" below means GREEN on that hypothesis row.
65
66 | Verdict | Trigger (pre-registered) | What we then do |
67 |---|---|---|
68 | **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3
H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha
Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the
single warmest |
69 | **PIVOT A ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3
GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing
useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the
product wedge, not the demo |
70 | **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and
liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever
owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |
71 | **PIVOT C ? payer is the parent, not the academy** | H5 RED from owners but H3
repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel
to parents) with the academy as channel, not customer |
72 | **FALSIFIED ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0
H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this
market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building
further |
73
74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide
75 explicitly with the owner ? but **the tallies above get computed first**, so the
76 discussion starts from the pre-registered read, not from anecdotes.
77
78 **Competitor-intel side-channel (always-on):** any visit where a competitor is
79 named ? fill the answer sheet's competitor block (what it does, what it misses,
80 what they pay). ?3 consistent reports about one product = update
81 `PMF_MARKET_RESEARCH.md` §4 with field-verified rows (today's table is web-sourced).
82
83 ## 4. Mechanics
84
85 - One [ `FIELD_VISIT_ANSWER_SHEET.md` ] (FIELD_VISIT_ANSWER_SHEET.md) per academy,
86 same day as the visit, in the associate's words + verbatim quotes.
87 - The owner (or Claude session) transcribes each sheet into one row of the rollup
88 table below (keep it in this doc or a sheet next to it); tallies and the
89 decision-rule read-out update per 5 visits.
90 - Demo asset if a conversation earns it: the real compiled reel
91 ( `output/MahadevpuraSingles_highlights.mp4` ) on a phone ? *show, never promise*
92 (flyer ground rule 3 still applies; the associate doesn't carry the demo,
93 the owner does on follow-ups).
94
95 ### Rollup table (one row per completed visit)
96
97 | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98 |---|---|---|---|---|---|---|---|---|---|---|---|
99 | 1 | | | | | | | | | | | |
100
101 ### Tally ? decision read-out (recompute per 5 visits)

```

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102
103 | Metric | Count | Threshold in play |
104 |---|---|---|
105 | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106 | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107 | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108 | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109 | H1 any recording | /n | <3 ? FALSIFIED leg |
110 | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112 ## 5. Guardrails that bind the field work
113
114 - **Phase 1 records nothing** ? notes only (flyer rule 7).
115 - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116   players; **sessions with minors are never recorded for our library** (training-pool
117   guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118 - Answer sheets contain personal contact details ? they stay between the associate
119   and the owner; transcribe to the rollup without phone numbers.
120

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4. user (2026-06-12T14:41:32.926Z)

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1      # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3      > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4      > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5      > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` $2.
6
7      ## Visit basics
8
9      | | |
10     |---|---|
11     | Date / time | |
12     | Academy name | |
13     | Area (neighbourhood) | |
14     | # courts in the hall | |
15     | Spoke to (name + role) | |
16     | Contact (phone/WhatsApp) ? *only if offered* | |
17     | How long the conversation ran | |
18
19     ## The seven questions ? their answers (verbatim where possible)
20
21     **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23     >
24
25     **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27     >
28
29     **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31     >
32
33     **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34     What's good, what's missing, what does it cost them?**
35
36     >
37
38     **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39     THEY say first (?)? What was their face doing?**
40
41     >
42
43     **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**

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44
45 >
46
47 **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48 to call)**
49
50 >
51
52 ## Required: the most negative thing they said (verbatim)
53
54 >
55
56 ## Best quotes (anything vivid, in their words)
57
58 >
59
60 ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score
lower)
61
62 | Hypothesis | Score |
63 |---|---|
64 | H1 recording behavior | GREEN / AMBER / RED |
65 | H2 dead footage | GREEN / AMBER / RED |
66 | H3 audience | GREEN / AMBER / RED |
67 | H4 whitespace | GREEN / AMBER / RED |
68 | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69 | H6 multi-court pain | GREEN / AMBER / RED |
70 | H7 Phase-2 door | GREEN / AMBER / RED |
71
72 ## Competitor block (only if they named one)
73
74 | | |
75 |---|---|
76 | Product | |
77 | What it does well (their words) | |
78 | What it misses (their words) | |
79 | What they pay | |
80
81 ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
82
83 ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
84
85 ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
really decides)
86
87 >
88

```

5. user (2026-06-12T14:41:33.903Z)

```

1 # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3 **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
academies are active)
4 **Location:** Bengaluru ? in-person visits to badminton academies
5 **Compensation:** ? ___ per completed academy visit + travel reimbursement *(discussed
individually)*
6 **Reports to:** Avi (phone/WhatsApp: ___ )
7
8 ---
9
10 ## The gig in one paragraph
11
12 We're building an AI tool that takes ordinary practice/match video from a badminton
academy and automatically finds

```

13 every rally, builds highlight reels, and generates match reports. Before building
further, we need honest, on-the-ground
14 answers from real academies about how they record, what they'd use, and what they'd pay.
**Your job: visit 10?15 good
15 academies, have a friendly 15?20 minute conversation with the owner/head coach using the
script below, and write down
16 what they actually say.** No selling. No tech knowledge needed. If a conversation goes
well, there's a Phase-2 follow-up
17 (coordinating a paid video-recording session) with additional compensation.
18
19 ---
20
21 ## Phase 1 ? The conversation (what you actually do)
22
23 **Who to talk to:** the academy owner or head coach ? the person who decides what the
academy spends money on.
24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is
available.
25
26 **Opening (keep it this honest):**
27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton
academies ? it takes normal practice
28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a
few serious academies to
29 > understand how you work with video today. Could I ask you a few questions? Takes 15
minutes, and we're not selling
30 > anything today."
31
32 **The questions (ask in this order; the starred ones matter most):**
33
34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired
person?
35 2. ? **Do you have recorded footage sitting around that nobody has time to watch?**
(multi-court hall recordings,
36 tournament days, CCTV) ? roughly how much?
37 3. Who would watch rally clips or highlights if they existed ? coaches for training?
players? parents?
38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory,
BadPro+, just YouTube/phone.)
39 What do you like / what's missing in it?
40 5. ? **If you could hand over a full session's video and get back every rally cut out +
a highlight reel + a simple
41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per
match?** (Let THEM say a number
42 first. Note their face, not just the words.)
43 6. Your hall has multiple courts running at once, right? Does that make video less
useful today?
44 7. Would you let us record one session here (we pay a small fee for the time) to show
you what the tool produces on
45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*
46
47 **After each visit, fill ONE answer sheet ? the template is
FIELD_VISIT_ANSWER_SHEET.md
48 (print one per visit):** academy name, location, # courts, person spoken to + role +
contact, answers 1?7 (their
49 words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a
50 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51 PMF_FIELD_SIGNALS.md ? that part is ours, not yours; just
capture honestly.)*
52
53 ---
54
55 ## Ground rules & gotchas (read twice ? this is the difference between useful and

```

useless)
56
57     1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58     figure out ? what would feel fair to you?" Never quote a price yourself.
59     2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60     what happened to that file?" gets the truth. Prefer the second kind of question.
61     3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62     (a paid recording session + free sample output) and only as "would you be open to
it".
63     4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64     than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65     5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66     doesn't, and what they pay. Spend extra time here.
67     6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68     and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69     7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70     the academy or players unless the owner offers.
71     8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72     9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73     don't share the answer sheets with anyone except Avi.
74
75     ---
76
77     ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79     You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80     one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81     record 1?2 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82     written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83     our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84     the tool).
85
86     ---
87
88     ## Interested?
89
90     Message Avi at ___ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91     walkthrough) so you're never improvising.
92

```

6. user (2026-06-12T14:41:34.906Z)

docs\PMF_FIELD_SIGNALS.md:70:| ****PIVOT B ? partner/API motion**** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

7. user (2026-06-12T14:41:43.701Z)

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1 # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3 > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4 > validate live before relying on them for investment/pricing decisions. This is the
5 > market-research-of-record behind the PMF section of the living
6 > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:
7 > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8 > self-recorded, **static-camera, full-court amateur** racket-sports video and
9 > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10 > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12 > **Confidence convention.** Figures from named industry reports, federations, or
13 > company pricing pages are tagged with source + year. Inferred items are tagged
14 > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15 > between vendors; ranges are shown rather than false precision.
16
17 ---
18
19 ## 1. Executive summary
20
21 The product sits at the intersection of three favorable trends: (a) a fast-growing
22 **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23 depending on vendor); (b) a structural shift to **amateur self-recording** on action
24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25 segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27 2024, +45.8% YoY).
28
29 The central tension is the **static-camera constraint**: the tool works best on fixed,
30 full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31 moat (most competitors target team sports / broadcast) and a filter on the addressable
32 user ? the customer must already set a camera on a tripod behind the court. That
33 behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34 the median casual player.
35
36 **Recommended beachhead (strongest single):** **badminton coaching academies and
37 serious club/competitive players in India** ? a large, rapidly formalizing academy
38 ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39 willingness to pay for coaching, an underserved tooling gap (incumbents are
40 team-sports/broadcast), and a direct-but-immature competitor field. The product is
41 *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
42 players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43 **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45 **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46 and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47 software-only BYO-camera model undercuts the hardware players; realistic anchors
48 **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50 **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51 coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52 variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53 China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55 ---
56
57 ## 2. Market sizing & trends
58
59 ### 2.1 Sports analytics / sports-video market
60 - **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61 Insights); other vendors USD 5.6?6.0B. **(2025)*
62 - **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). **(reports 2024?2025)*
65 - **Video analytics is the leading segment** within sports analytics (Grand View /

```


Fortune BI) ? the directly relevant sub-market.

- **Interpretation (estimate):** the **amateur/club auto-highlight** slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- **2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). **(2024?2025)**
- **Sports is the largest end-use segment** of action cameras (GMIInsights, 2024).
- **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:** the capture-side behavior the product depends on is already widespread and growing ? **for the segments that record** ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	Badminton ? China ~80M registered, up to ~100M (largest base) industry compilations (2023)
Badminton	India 20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	Badminton ? Indonesia National sport; ~1M club players industry (est.)
Badminton	Denmark 100,000+ registered club members (Europe #1) industry (2023)	Badminton ? Japan/Korea Major school-club + workplace sport industry (2023)
Tennis	106M players (2024), +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	Tennis ? Asia ~1/3 of players (33M of 87M) ITF (2021)
Pickleball	US 19.8M players (2024), +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	Pickleball ? US courts ~70,000 courts, +55% places to play YoY SFIA / Pickleheads (2024)
Pickleball	global IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim (low confidence) various (2024?2025)	Table tennis 30M+ players; 227 member associations ITTF (current)
Cricket	2.5B+ fans; 500M+ in India (mostly fans , not self-recorders) Cricket Rise / Nielsen (2024?2025)	

- **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR (Market.us/Custom Market Insights, 2024?2025).
- **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

Geographic read: badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as **fandom** but its amateur self-recording + static-full-court fit is weak (large field, not a fixed-frame court). For **this product's static-camera fit**, badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.

3. Customer segment evaluation

Segment	Market-size signal	Willingness to pay	Accessibility	Static-camera fit
B2C/B2B	Verdict			
Individual amateur/enthusiast	Largest raw TAM (100s of millions) Low ? competes with free; ~USD 5?15/mo ceiling High via app stores; high CAC/churn Mixed ? most won't set a tripod B2C Volume play, weak economics; use as funnel			
Coaches & coaching academies	India academy boom; thousands across Asia High ? already sell coaching; USD 300?1,500/yr plausible Reachable via federations/distributors/social Strong ? already film fixed-angle B2B/B2B2C Best fit ? buyer, recorder, payer aligned			

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary; longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor ? panning = failure mode**** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/

124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest

125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters

126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

132 | Product | Category | Model | Price | Notes (year) |

133 |---|---|---|---|---|

134 | ****Veo (Cam 3)**** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300 (Pro) | Camera leased free with sub (2025) |

136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD 69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ? 6,100/yr (club) | Institutional budgets (2025) |

138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports incumbent (2026) |

139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD 179.99/yr****; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |

140 | ****BadPro+**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "< one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025) |

141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true auto-highlight tools (2025) |

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD

144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,

145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?

146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +

147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and

148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B

149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline

150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****

151 value story more than a B2C one.

152

153 ---

154

155 **## 5. Recommended beachhead**

156

157 Ranked by ****WTP × participation density × existing recording behavior × static-camera**

158 **fit × competitive whitespace.****

159

160 **### #1 (strongest) ? Badminton coaching academies & serious players, ****India******

161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo

162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize

163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton

164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.

165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not

166 the individual.

167

168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**

169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;

170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**

171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+

172 monthly users) ? head-to-head differentiation, not greenfield.

173

174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,

175 Korea) + Denmark**

176 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea

177 school-club & workplace systems), mature club infra, fixed-court recording.

178 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia

179 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer

181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C

182 motion**, then expand to **pickleball-US** (out-build SwingVision) and

183 **badminton-Asia/Denmark** (density).

184

185 ---

186

187 ## 6. Risks & open questions (what would invalidate the thesis)

188

189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a

190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +

191 50?100 enthusiasts on whether they already record fixed-angle.

192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap

193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.

194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,

195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real

196 amateur clips; define a "good-capture" spec + auto-reject (ties to

197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).

198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are

199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy

200 INR licensing.

201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).

202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match

203 (offline self-improving detection, per-rally tagging).

204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/

205 academy self-serve before institutional deals.

206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real

207 purchasing criterion via 5?10 buyer interviews.

208

209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy

210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door

211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

212

213 ---

214

215 ## 7. Sources

216

217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View

218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*

219 - <https://www.precedenceresearch.com/sports-analytics-market>

220 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>

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222 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>

223 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>

224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)

225 [market](https://www.gminsights.com/industry-analysis/action-camera-market) (2024);

226 <https://www.precedenceresearch.com/action-camera-market> (2024);

227 <https://electroi.com/stats/action-camera-statistics/> (2024)

228 - Badminton participation/market: [https://badmintonbites.com/how-many-badminton-players-](https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/)

are-there-in-the-world/ ;

<https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its->

popular/ ;
229 <https://en.wikipedia.org/wiki/Badminton> ;
230 <https://www.marketgrowthreports.com/market-reports/badminton-market-110049> ;
231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
232 - Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
233 <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
234 - Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
235 <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
241 - Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing>
;
242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
243 <https://www.badproplusapp.com/>
244

8. user (2026-06-12T14:41:44.936Z)

c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md:388: subs rev \$2.75M +12% YoY; PB Vision \$99/yr + **\$8/hr-of-video partner API** ; Gemini COGS ~\$0.30?0.60/hr ? fat margin);
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md:394: (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) + **Gemini-wrapper baseline vs the

9. user (2026-06-12T14:41:55.776Z)

docs\BACKLOG.md:8:_Last updated: 2026-06-10._
docs\BACKLOG.md:17:## Engineering hardening deferred from the 2026-06-10 audit sweep
docs\CODE_MAP.md:7:[Omitted long matching line]
docs\CODE_MAP.md:142:[Omitted long matching line]
docs\HOSTING_PLAN.md:4:**Date:** 2026-06-10 · **Constraints:** budget \$10?25/mo · GCP credits available · DigitalOcean
docs\mockups\ui_alpha_mockup.html:172:<div class="mock">static mockup · fake data · docs/mockups · 2026-06-10</div>
docs\PMF_FIELD_SIGNALS.md:31:hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
docs\PMF_FIELD_SIGNALS.md:70:| **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision × PodPlay template from the 2026-06-10 audit) |
docs\NEXT_STEPS.md:3:**Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
docs\NEXT_STEPS.md:36:Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).
docs\NEXT_STEPS.md:64:6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
docs\archives\decisions\DECISIONS.md:390:**Date:** 2026-06-10
docs\archives\decisions\DECISIONS.md:414: construction ? the fps-normalization bug (fixed 2026-06-10) is the skew class this kills.
docs\archives\decisions\DECISIONS.md:469:2026-06-10.
docs\archives\decisions\DECISIONS.md:473:**Date:** 2026-06-10 · **Status:** RATIFIED (owner, in-session)
docs\UI_ALPHA_EXECUTION_PLAN.md:3:**Status:** ACTIVE · updated 2026-06-10 · **Linked:**
`UI\_PROPOSAL.md` · `HOSTING\_PLAN.md` ·

docs\UI_ALPHA_EXECUTION_PLAN.md:12:| 2026-06-10 | Product name ****KhelSutra Guru**** . domain ****khelsutra.guru**** (GoDaddy, 3-yr + protection) |

docs\UI_ALPHA_EXECUTION_PLAN.md:13:| 2026-06-10 | Separate ****private repo `khelsutra`**** ("heart and brain"): `web/` SPA now, control plane at beta; backend enablers stay in the indexer repo |

docs\UI_ALPHA_EXECUTION_PLAN.md:14:| 2026-06-10 | ****Local-first delivery**** (this plan's phase structure) |

docs\UI_ALPHA_EXECUTION_PLAN.md:22:- [x] ****Owner:**** git-init + private GitHub push ? done 2026-06-10:

docs\UI_ALPHA_EXECUTION_PLAN.md:24:- [x] ****Owner:**** explicit "go" for PR-1 ? given 2026-06-10

docs\UI_ALPHA_EXECUTION_PLAN.md:25:- [x] ****Owner:**** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key deletion in AI Studio)

docs\UI_ALPHA_EXECUTION_PLAN.md:68:`ai\_max\_workers=1`, 6-attempt backoff ? learned 2026-06-10) . CF free-tier 100 MB body cap

docs\UI_PROPOSAL.md:4:****Date:**** 2026-06-10 . ****Inputs:**** PMF_MARKET_RESEARCH, COMMERCIALIZATION, PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES, reporting/spec.yaml, current API + static UI.

docs\UI_PROPOSAL.md:16:Serving economics per the 2026-06-10 quality table: alpha serves the ****Gemini path****

docs\UI_PROPOSAL.md:98:2. ****RESOLVED 2026-06-10:**** product name = ****KhelSutra Guru****; domain = ****khelsutra.guru**** (GoDaddy receipt, 3-yr registration + domain protection, renews 2029-06).

docs\UI_PROPOSAL.md:102:6. ****RESOLVED 2026-06-10:**** separate ****private repo `khelsutra`**** (owner-ratified ? the product's

docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:35:5. Packaging modernization (`pyproject.toml` + clean entry points). ? ****NOT shipped**** (no `pyproject.toml` on master as of 2026-06-10 ? this entry was aspirational; corrected here per attribution-honesty). Still tracked as COMMERCIALIZATION C9 / DEFERRED_HARDENING_PLAN #7.

10. user (2026-06-12T14:41:57.189Z)

37-| C12 | ****SaaS foundations (async/auth/tenancy)**** | ****P1**** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM\_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED\_HARDENING\_PLAN.md` (#16). |

38-| | ****? IN-FLIGHT (in progress / watch)**** | | | |

39-| C4 | ****Runtime AI strategy (A/B/C + owned model)**** | ****P0**** | ? | Live = ****C**** Gemini 2.5 Flash, paid tier. Long-term = ****own a self-hosted generative video-QA model**** (fine-tuned open VLM) as primary, ****paid Gemini always-ready as fallback**** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model ****after**** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |

40:| C13 | ****Beachhead validation**** | ****P0**** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends ****India badminton academies**** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |

41-| C8 | ****Golden-set vendoring spend**** | ****P1**** | ? | ADR-006: ****Roora (Bengaluru)**** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |

42-[Omitted long context line]

43-[Omitted long context line]

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201-## Changelog

202-

203-- ****2026-06-07**** ? Added the market-research-of-record (`PMF\_MARKET\_RESEARCH.md`):

204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13

205: "beachhead validation" workstream. Wired the summary into §6.

206-- ****2026-06-07**** ? Created this living tracker. Merged the 2026-05-30 monetization

207- audit + the commercial-readiness review. ****Refreshed since the audit:**** C1 AGPL

11. user (2026-06-12T14:41:57.391Z)

1 # Next steps (live) ? owned-model + multisport

2

3 ****Updated:**** 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

4 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-

```

first delivery ? see
5     `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
```

**PR-2

```

6     upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
7     compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
8     coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
9     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
10    **Authoritative roadmap:**
11    `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
12    `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
13
14    ## Where we are (one paragraph)
15    ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
16    behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
17    the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
18    eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
19    sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
20    (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
21    battery is done** (see table) ? the moat is quantified.
22
23    ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
24    | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
25    |---|---|---|
26    | Heuristic (velocity windowing) | 0.657 | 0.157 |
27    | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
28    | Two-stage WASB+Gemini refine | 0.83 | ? |
29    | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
30
31    **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
32    wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
33    the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
34    architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
35    only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
36    Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
37
38    ## Prioritized next steps
39    1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
40    Priority order for new videos: ** (a) multi-court badminton** (the target distribution
AND where the ship target
41    lives), ** (b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
42    corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
43    `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
44    label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
```

```

re-run the Gen-0 harness.
45     The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
46     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
47         reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
48         (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
49         paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
50     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51         2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52         calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54         badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55         by C3 (provenance multiplies per sport).
56     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
57         "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
58         (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
59         `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
60         physical-visit approach):** `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (recruit hand-
out: script, ground rules,
61         Phase-2 consent; owner fills ?/contact placeholders) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture
62         form) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R scoring anchors + **pre-registered
decision rules** ? PROCEED /
63         PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
64     6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
65         deletion in AI Studio); **rename repo off "badminton"** (owner action,
66         in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
67
68     ## Pointers / artifacts
69     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
70     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
71     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
72     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
73     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
74     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
75     (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
76     labels: `...\Collect\Golden Labelled\`.
77     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
78     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
79     with a MISLEADING "credits depleted" 429; ramp gradually.
80     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
81     PLATFORM slice = async job model (single-tenant).
82

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12. user (2026-06-12T14:42:07.458Z)

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360    ?? Audit's **coverage-inversion** finding (wasb_hybrid/validators untested) remains open
? CI only guards the
361    import/syntax class + whatever the suite covers.
362
363    **Checkpoint:** 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit
`wf_2a64bc88-2a8` over indexer+collector:
364    engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors
completed+verified, 5 rate-limited).
365    - ? **CRITICAL ? master is BROKEN since 7fbbeb0 (2026-06-07 print?logger sweep):**
366    `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines
(100/104/156/160/181/191/213)** ? SyntaxError ?
367    `import backend.main` fails ? FastAPI server + `/api/compile` dead. Triple-verified +
reproduced first-hand
368    (`py_compile` exit 1). The **254-green suite never noticed** (zero tests import
backend.main / compiler / wasb_hybrid /
369    motion / any concrete validator; **no CI exists at all**). Spawned fix chip
`task_2b7a19d4` (fix + import/compileall
370    smoke test). Coverage is INVERTED vs production risk (tracknet twin 8 tests vs live
wasb_hybrid 0; TrackNetRunner 21
371    vs WasbRunner 1). Suite itself healthy where it exists (254 green / 19.9s / faithful
mocks / low flake risk); "fully
372    mocked runs anywhere" is false on cv2-free machines (bare `import cv2` at collection
time, only obsreport importorskips).
373    - **Doc drift (verified, one-directional ? newest decisions not back-propagated):** (1)
licensing "Apache-2.0 only" still
374    in CLAUDE.md:33 / PLATFORM:269,301 / COMMERCIALIZATION C14 vs ratified
**MIT/Apache/BSD** (a literal CLAUDE.md reader
375    rejects the plan's own MIT ActionFormer/ASFormer heads); (2) **Gemma-4 still
"approved" in those same docs vs
376    AMBER/bench-only in the plan; (3) **M0.3 purge pointer WRONG** ?
`training.label_candidates` is a pure GT-agnostic fn;
377    the live Gemini-silver training violation = `backend/eval/distill_local.py` (+
calibrate_local thresholds) + gitignored
378    `output/local_rally_model.json`; (4) **C3 escape hatch broken** ? "train-own weights
(Phase-4 roadmap)" cites a phase
379    that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T
train a detector) ? own-weights is
380    an unscoped new workstream (coordinate labels or detector swap); (5)
OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are
381    linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS
is branch-only; (6) README
382    "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default";
(7) ROADMAP still says audio
383    "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve";
(8) Roora multisport CSVs live ONLY
384    in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline
TT-0.909/pickleball-0.308 evidence is one cleanup from
385    loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per
sport"); (10) ship-gate F1@tIoU still
386    undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no
tracker (BACKLOG lacks it).
387    - **PMF (web-grounded, sources in audit output):** pricing/cost benchmarks VERIFIED
(SwingVision $179.99/yr ~20k paying
388    subs rev $2.75M +12% YoY; PB Vision $99/yr + **$8/hr-of-video partner API**; Gemini
COGS ~$0.30?0.60/hr ? fat margin);
389    but **"India badminton whitespace" FALSIFIED ? Bangalore is the most contested
badminton-AI city**: Machaxi ($1.5M
390    Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School),
Game Theory (Gopichand-backed smart
391    courts w/ auto-highlights), Stupa ($3.3M, BWF-approved, TT?badminton) ? NONE in our
2026-06-07 competitive table.
392    Surviving wedge (narrow, real): **software-only rally-precise indexing of EXISTING
messy multi-court BYO footage**
393    (market solves it only with per-court hardware) sold per-hour as engine/API, possibly
THROUGH those players

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394      (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) +
**Gemini-wrapper baseline vs the
395      n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid
academy pilot.
396      - **AUDIT NOW COMPLETE (8/8 auditors, resumed `wa4d2980l` after Max-20x refresh; 40
agents).** New high/critical findings
397      beyond the first batch:
398      - ? **CRITICAL (collector) ? LICENSING GUARDRAIL VIOLATION:** `src/cli/ingest.py:12`
defaults `--license MIT`; the
399      ShuttleSet **annotation** license is stamped onto **YouTube broadcast video** (e.g.
shuttleaset_21 = a YONEX Thailand
400      Open broadcast, `is_commercial=true, license_type=MIT`). **24 videos / 1915 segments
currently exported as
401      "commercially-safe" on unsound grounds** ? directly violates the "training data must
be commercial-clean" guardrail.
402      - **HIGH (indexer, all verified):** (a) **nested config sections silently drop unknown
keys** ? only `AppConfig` has
403      `extra='allow'`, so the ENTIRE `indexing.wasb` block
(weights/conda_env/velocity_threshold) + `yolo_prefetch_workers`
404      + `debug_duration` are unconfigurable via config.json; meanwhile
`yolo_parallel_chunks/yolo_chunk_workers` are read
405      nowhere. (b) **detector artifacts keyed by file BASENAME not video_id** ? two
`match.mp4` files reuse stale frames ?
406      a trajectory + candidate_segments computed from the WRONG video (silent training-
data poisoning). (c) **re-ingest is
407      destroy-before-confirm** ? `add_video('processed'/'failed')` INSERT-OR-REPLACE
cascade-deletes prior good segments
408      BEFORE the new run; a failed/empty re-run leaves zero. (d) **stitching padding
config not wired** ? API reels always
409      end_padding=1.0 despite config 0.5. (e) **`/api/process` arbitrary server-file read
+ `/api/compile` path-traversal/

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13. user (2026-06-12T14:42:09.126Z)

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86:  shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
149:  local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.

```

14. user (2026-06-12T14:43:34.347Z)

Structured output provided successfully

15. assistant (2026-06-12T14:43:40.473Z)

Verification complete. The claim is real but downgraded to low severity: the dangling "PB Vision × PodPlay" citation in PIVOT B is genuinely unresolvable in-repo (verified by repo-wide grep), but the pivot row's trigger and operative action are self-contained, and the doc's intended readers (owner + Claude sessions) can resolve it via session memory.